Assessment Details

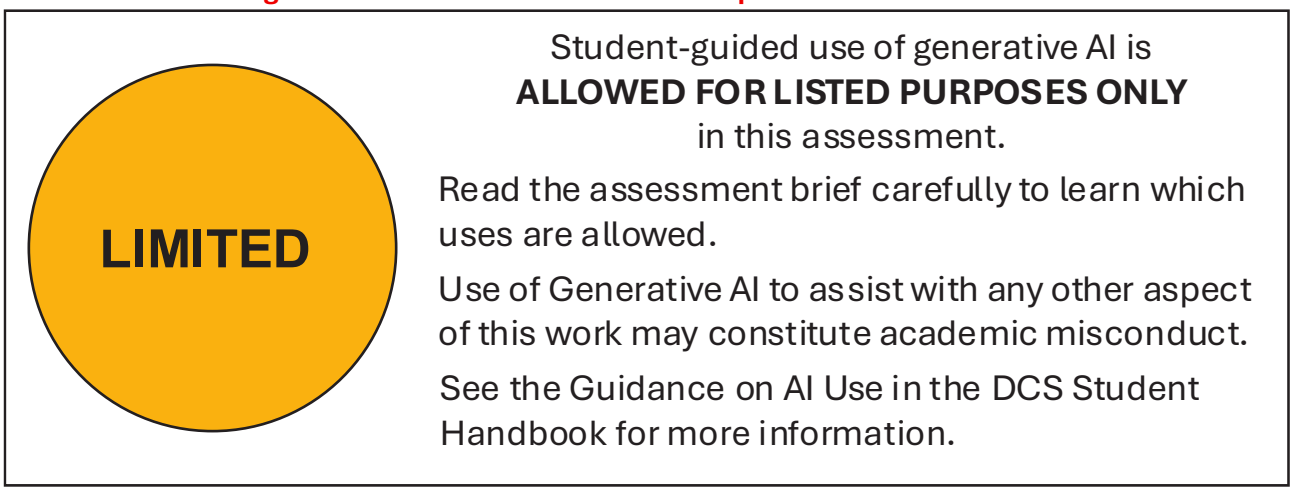
CS342 – Coursework

UNIVERSITY OF WARWICK | Department of Computer Science

General Assessment Information

Module Code	**CS342**
Module Title	**Machine Learning**
Module Organiser(s)	**Shan - Shan.Raza@warwick.ac.uk** **Long - Long.Tran-Thanh@warwick.ac.uk**
How to get in touch	**Shan - Please contact via email at Shan.Raza@warwick.ac.uk** **Long - Please contact via email at Long.Tran-Thanh@warwick.ac.uk**
Assessment Title	**Coursework**
Assessment Weighting / Credits	**40%**
Submission Deadline	**12pm – Term 1 Week 11 - Tuesday (2026-12-15)**

Use of Generative AI: Generative AI tools are not permitted for Task 1. However, these are permitted as specified for remaining tasks. Please see each task description for further details.



Eligible for Self-Certification	**No**
Feedback Format	Students will receive an overall mark for the coursework, together with a breakdown of marks for each task: **Task 1: Exploratory Data Analysis and Visualisation (20 marks)** **Task 2: Unsupervised Classic Machine Learning (20 marks)** **Task 3: Supervised Classic Machine Learning (20 marks)** **Task 4: Unsupervised Deep Learning (20 marks)** **Task 5: Supervised Deep Learning (20 marks)** Feedback will indicate how marks were awarded against the published marking criteria and identify areas of strength and areas for improvement. In addition, students will receive a short paragraph of personalised feedback commenting on their data analysis, model development, evaluation, and interpretation of results. Where appropriate, suggestions for improvement will be provided. General feedback will also be released to the whole cohort, highlighting common strengths, areas for improvement, and examples of good practice observed across submissions.

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Submission Details

Method	• Online submission via Tabula You must also submit an AI declaration at the submission portal. AI Declaration Form: https://warwick.ac.uk/fac/sci/dcs/teaching/forms/ai-declaration/
Format of submission	PDF (Report) and Jupyter Notebooks
Submission length	2000 words report
Formatting instructions	Font Size - Arial 12pts, single spacing
Additional Information	Do not include input data as part of your submission. You can highlight cells to read the input data in your Jupyter Notebooks.

Intended Learning Outcomes (ILOs) Assessed

Learning Outcome	How the assessment links to the Learning Outcome
Develop an appreciation for what is involved in models that are optimised based on data	The assessment requires students to develop understanding of the input (battery life) data. The students need to demonstrate the choice of models used to process the data.
Understand a wide variety of learning algorithms	The students need to identify and implement an appropriate machine learning algorithm which can perform unsupervised and supervised classification tasks using classic machine learning and deep learning.
Understand how to evaluate models optimised on data	Using the trained models, the students need to demonstrate their understanding of how the models performed. Why one model performed better than the other and if they can identify features which can be used to improve the performance of these models.
Understand applications of models to real data and problems	The battery life dataset is chosen for this assessment to help the students understand applications of machine learning to real life datasets.

Referencing & Avoiding Academic Misconduct

It is serious Academic Misconduct to pass off the work of others (including peers or AI-generated content) as your own. You should not permit colleagues to copy from you.

Sources must be appropriately and properly acknowledged every time reference is made to another's work, using the Harvard or Vancouver Referencing system. Failure to do so amounts to plagiarism which breaches university regulations and falls short of the Academic Integrity expected in the department and university.

Find out more about referencing from the Library: https://warwick.ac.uk/services/library/students/referencing/referencing-styles/

Reminder: It is your responsibility to be familiar with the department's academic misconduct policy, and its guidance around the use of generative AI tools.

DCS AI Policy (Student Handbook): https://warwick.ac.uk/fac/sci/dcs/teaching/handbook/examinations/#aitools

DCS Academic Misconduct Policy (Student Handbook): https://warwick.ac.uk/fac/sci/dcs/teaching/handbook/examinations/#academicmisconduct

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Assignment Brief: Predicting Battery Useful Life

Introduction

Lithium-ion batteries power electric vehicles, renewable energy systems, and a wide range of consumer devices. As batteries age, their performance degrades and their risk of failure increases. Accurate analysis of battery degradation data is critical for predictive maintenance, reliability engineering, asset management, battery repurposing and recycling.

Recent advances in machine learning have enabled data-driven approaches for battery prognostics. By analysing historical battery measurements, machine learning models can discover hidden patterns in battery behaviour, identify groups of batteries exhibiting similar degradation characteristics, and predict future battery performance. These capabilities support more informed maintenance decisions and improve asset management strategies.
In this coursework, you will apply both unsupervised learning and supervised learning techniques to analyse a large-scale battery degradation dataset. This dataset has been downloaded from Kaggle (https://www.kaggle.com/datasets/razanihababdellatif/battery-remaining-useful-life-rul-dataset/data).
Feel free to read through example codes on Kaggle but you are **not allowed** to copy solutions from the webpage.

We have uploaded the data for this coursework to ` /modules/cs342/2026/coursework/data/`.

You should have access to the following files:

1. battery_metadata.csv
2. battery_observations.csv
3. train_batteries.csv
4. validation_batteries.csv
5. test_batteries.csv

In this coursework, we would only focus on the classification task i.e., to predict the **degradation_status** category ('Normal', 'Moderate', 'Severe', 'Critical') of a lithium-ion battery. Therefore, the **output variable** for the learning task is: **degradation_status**

For all the tasks, students may use information from:

battery_metadata.csv
battery_observations.csv

However, the following variables must **not** be used in any of the following tasks as these are related to future battery health or degradation outcomes:

remaining_useful_cycles
remaining_useful_efc
state_of_health_pct

The dataset includes predefined splits for training (**train_batteries.csv**), validation (**validation_batteries.csv**) & test (**test_batteries.csv**). Students must use these partitions as supplied and must not create alternative train-test splits.

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Tasks

Task 1: Exploratory Data Analysis and Data Visualisation (20 Marks)

Before developing machine learning models, it is important to develop an understanding of the dataset and identify factors that may influence battery degradation and useful life.

The task is to conduct an exploratory analysis of the battery dataset using appropriate data visualisation techniques. You should investigate the characteristics of the data, identify patterns and relationships between variables, and identify features that may be useful for predicting battery degradation and remaining useful life. Visualisations should be accompanied by clear interpretation and discussion of your findings. Suitable tools include Matplotlib and Seaborn, although other visualisation libraries may also be used. Students are expected to independently select, create, and interpret visualisations using their own judgement.

Generative AI tools are <u>not</u> permitted for this task.

- Quality and appropriateness of visualisations: **6 marks**
  - 5-6 marks (Excellent): Visualisations are well chosen, clearly presented, and highly relevant to the investigation. A range of appropriate visualisation techniques is used effectively to explore the data and support conclusions.
  - 3-4 marks (Good/Satisfactory): Visualisations are generally appropriate and correctly produced, although some may provide limited insight or there may be opportunities to use more suitable techniques.
  - 1-2 marks (Limited): Visualisations are basic, poorly chosen, difficult to interpret, or provide little evidence of thoughtful exploration of the dataset.
  - 0 marks: Little or no meaningful visualisation provided.
- Interpretation and insight from the analysis: **7 marks**
  - 6-7 marks (Excellent): Demonstrates strong understanding of the dataset. Patterns, trends, anomalies, and relationships are accurately identified and supported by evidence from the visualisations. Discussion is insightful and well justified.
  - 3-5 marks (Good/Satisfactory): Provides reasonable interpretation of the visualisations and identifies some relevant patterns and relationships, although discussion may be descriptive rather than analytical.
  - 1-2 marks (Limited): Interpretation is superficial, inaccurate, or largely descriptive with little evidence of critical analysis.
  - 0 marks: Little or no meaningful interpretation provided.
- Identification and justification of useful features: **7 marks**
  - 6-7 marks (Excellent): Clearly identifies potentially useful features for predicting battery degradation status and provides strong evidence and justification based on the exploratory analysis. Demonstrates critical consideration of feature relevance.
  - 3-5 marks (Good/Satisfactory): Identifies several potentially useful features with some justification from the analysis, although the reasoning may be incomplete or only partially supported.
  - 1-2 marks (Limited): Feature selection is poorly justified, based on weak evidence, or shows limited understanding of the relationship between the features and the prediction task.
  - 0 marks: No meaningful attempt to identify or justify useful features.

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Task 2: Unsupervised (Classic) Machine Learning for Battery Degradation Analysis (20 Marks)

Develop and apply an unsupervised machine learning model (non-neural network) to investigate patterns within the battery dataset.

You should explain how your chosen method was designed, including any data preprocessing, feature selection, and parameter choices. Evaluate whether the resulting output corresponds to meaningful battery degradation groups and discuss whether the features identified in Task 1 were useful to train the model. Appropriate visualisations should be used to support your discussion.

You must not use **remaining_useful_cycles, remaining_useful_efc or state_of_health_pct** as input (or output) to train your models.

Generative AI tools may be used for this task, provided their use is appropriately acknowledged in the report.

- Selection and implementation of an appropriate unsupervised learning method: **6 marks**
  - 5-6 marks (Excellent): An appropriate unsupervised learning method is selected, correctly implemented, justified, and applied effectively to the problem. The methodology is technically sound and produces meaningful results.
  - 3-4 marks (Good/Satisfactory): An appropriate method is selected and implemented correctly, although there may be minor technical issues or limitations in its application.
  - 1-2 marks (Limited): The method selected is inappropriate for the task, poorly implemented, or produces results that are difficult to interpret.
  - 0 marks: No meaningful implementation provided.
- Justification of preprocessing, feature selection and parameter choices: **6 marks**
  - 5-6 marks (Excellent): Clear and well-reasoned justification is provided for preprocessing steps, feature selection decisions, and model parameters. Choices are supported using evidence from the data and findings from Task 1.
  - 3-4 marks (Good/Satisfactory): Reasonable justification is provided for most choices, although some decisions may lack supporting evidence or discussion.
  - 1-2 marks (Limited): Little justification is provided, or decisions appear arbitrary and unsupported.
  - 0 marks: No meaningful justification provided.
- Analysis and interpretation of results: **5 marks**
  - 4-5 marks (Excellent): Results are critically analysed and interpreted. The student demonstrates a strong understanding of the discovered patterns and evaluates whether the resulting groups correspond to meaningful battery degradation behaviour.
  - 2-3 marks (Good/Satisfactory): Results are described and interpreted correctly, although the discussion may be largely descriptive or lack depth.
  - 1 mark (Limited): Minimal interpretation is provided, or conclusions are poorly supported by the results.
  - 0 marks: No meaningful analysis provided.
- Discussion of the usefulness of features identified in Task 1: **3 marks**
  - 3 marks (Excellent): Provides a clear discussion of how features identified in Task 1 contributed to model performance and pattern discovery, supported by evidence from the analysis.
  - 2 marks (Satisfactory): Makes some connection between Task 1 findings and the unsupervised analysis, although the discussion is limited.
  - 1 mark (Limited): Only a superficial discussion of feature usefulness.
  - 0 marks: No discussion provided.

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Task 3: Supervised (Classic) Machine Learning for Battery Degradation Analysis (20 Marks)

Develop and evaluate a supervised machine learning model (non-neural network) to predict the battery **degradation_status** category (Normal, Moderate, Severe, or Critical).

You should describe and justify your modelling approach, including any preprocessing, feature engineering, feature selection, and hyperparameter tuning. Evaluate the model using appropriate classification metrics and compare its performance with the unsupervised approach developed in Task 2. Discuss whether supervised learning improves classification performance and explain why this may be the case. Reflect on whether the features identified in Tasks 1 and 2 were useful for classification and identify the most important features contributing to model performance.

You must not use **remaining_useful_cycles, remaining_useful_efc or state_of_health_pct** as input (or output) to train your models.

Generative AI tools may be used for this task, provided their use is appropriately acknowledged in the report.

- Selection and implementation of an appropriate supervised learning model: **6 marks**
  - 5–6 marks (Excellent): An appropriate supervised learning method is selected, correctly implemented, and trained effectively. The model is technically sound and demonstrates strong predictive performance.
  - 3–4 marks (Good/Satisfactory): An appropriate model is selected and implemented correctly, although there may be minor issues in implementation, tuning, or model performance.
  - 1–2 marks (Limited): The model selected is inappropriate, poorly implemented, or fails to produce meaningful classification results.
  - 0 marks: No meaningful implementation provided.
- Justification of preprocessing, feature selection, and model design: **6 marks**
  - 5–6 marks (Excellent): Clear and well-supported justification is provided for preprocessing steps, feature engineering, feature selection, model design, and hyperparameter choices. Decisions are supported by evidence from Tasks 1 and 2.
  - 3–4 marks (Good/Satisfactory):Reasonable justification is provided for most modelling decisions, although some choices may lack sufficient explanation or supporting evidence.
  - 1–2 marks (Limited): Little justification is provided, or modelling decisions appear arbitrary and unsupported.
  - 0 marks: No meaningful justification provided.
- Evaluation and interpretation of classification performance: **5 marks**
  - 4–5 marks (Excellent): Appropriate evaluation metrics are used and interpreted correctly. Performance is critically analysed, strengths and limitations are identified, and conclusions are supported by evidence.
  - 2–3 marks (Good/Satisfactory): Performance is evaluated using appropriate metrics and interpreted reasonably well, although the discussion may be largely descriptive.
  - 1 mark (Limited): Minimal evaluation is provided or performance metrics are interpreted incorrectly.
  - 0 marks: No meaningful evaluation provided.
- Comparison with Task 1 & Task 2 and discussion of feature importance: **3 marks**
  - 3 marks (Excellent): Provides a clear comparison with findings from Tasks 1 and 2, discusses whether supervised learning improves performance, and presents a well-supported discussion of feature importance.
  - 2 marks (Satisfactory): Makes reasonable comparisons with previous tasks and identifies some important features, although the discussion may lack depth.

- 1 mark (Limited): Only a superficial comparison is provided with limited discussion of feature importance.
- 0 marks: No meaningful comparison or discussion provided.

Task 4: Unsupervised Deep Learning for Battery Degradation Analysis (20 Marks)

Develop and apply an unsupervised deep learning model to investigate patterns within the battery dataset.

You should explain how the model was designed, including any preprocessing, network architecture, latent space representation, and parameter choices. Evaluate whether the learned representations reveal meaningful battery degradation groups and discuss whether the features identified in Task 1 were useful for the analysis. Appropriate visualisations should be used to support your discussion.

You must not use **remaining_useful_cycles, remaining_useful_efc or state_of_health_pct** as input (or output) to train your models.

Generative AI tools may be used for this task, provided their use is appropriately acknowledged in the report.

- Selection and implementation of an appropriate unsupervised deep learning model: **6 marks**
  - 5–6 marks (Excellent): An appropriate unsupervised deep learning approach is selected, correctly implemented, and applied effectively to the dataset. The model architecture is suitable for the problem and produces meaningful latent representations.
  - 3–4 marks (Good/Satisfactory): An appropriate model is selected and implemented correctly, although there may be minor issues in architecture design, training, or application.
  - 1–2 marks (Limited): The model selected is inappropriate, poorly implemented, or produces representations that are difficult to interpret or justify.
  - 0 marks: No meaningful implementation provided.
- Justification of preprocessing, feature selection, and model design: **6 marks**
  - 5–6 marks (Excellent): Provides clear and well-supported justification for preprocessing, feature selection, architecture design, latent space size, training procedure, and parameter choices. Decisions are supported by evidence from the data and findings from previous tasks.
  - 3–4 marks (Good/Satisfactory): Provides reasonable justification for most design choices, although some decisions may lack supporting evidence or discussion.
  - 1–2 marks (Limited): Limited or weak justification is provided, with several design choices appearing arbitrary or unsupported.
  - 0 marks: No meaningful justification provided.
- Analysis and interpretation of the learned representations: **5 marks**
  - 4–5 marks (Excellent): Provides insightful analysis of the learned representations and demonstrates a strong understanding of the underlying patterns captured by the model. Appropriate visualisations are used effectively to support conclusions about battery degradation groups.
  - 2–3 marks (Good/Satisfactory): Provides reasonable interpretation of the learned representations and identifies some meaningful patterns, although the discussion may be largely descriptive.
  - 1 mark (Limited): Minimal interpretation is provided, or conclusions are weakly supported by the results.
  - 0 marks: No meaningful analysis provided.
- Discussion of the usefulness of features identified in Task 1, 2 & 3: **3 marks**

- 3 marks (Excellent): Provides a clear and evidence-based discussion of how features identified in previous tasks contributed to the learned representations and pattern discovery. Makes meaningful connections between the findings of Tasks 1–4.
- 2 marks (Satisfactory): Makes some connection between previous findings and the deep learning analysis, although discussion may lack depth or supporting evidence.
- 1 mark (Limited): Provides only a superficial discussion of feature usefulness.
- 0 marks: No meaningful discussion provided.

Task 5: Supervised Deep Learning for Battery Degradation Analysis (20 Marks)

Develop and apply an supervised deep learning model to investigate patterns within the battery dataset.

You should describe and justify your modelling approach, including any preprocessing, feature engineering, architecture selection, and hyperparameter tuning. Evaluate the model using appropriate classification metrics and compare its performance with the supervised machine learning model developed in Task 3. Discuss whether deep learning improves classification performance and explain why this may be the case. Reflect on whether the features identified in Tasks 1, 2, and 4 contributed to the final model performance.

You must not use **remaining_useful_cycles, remaining_useful_efc or state_of_health_pct** as input (or output) to train your models.

Generative AI tools <u>may be used</u> for this task, provided their use is appropriately acknowledged in the report.

- Selection and implementation of an appropriate deep learning model: **6 marks**
  - 5–6 marks (Excellent): An appropriate deep learning model is selected, correctly implemented, and trained effectively. The architecture is well suited to the problem, demonstrates strong predictive performance, and is implemented to a high technical standard.
  - 3–4 marks (Good/Satisfactory): An appropriate model is selected and implemented correctly, although there may be limitations in architecture design, training, or overall performance.
  - 1–2 marks (Limited): The model selected is inappropriate for the task, poorly implemented, or produces weak or unreliable classification results.
  - 0 marks: No meaningful implementation provided.
- Justification of preprocessing, architecture, and training strategy: **5 marks**
  - 4–5 marks (Excellent): Clear and well-supported justification is provided for preprocessing, feature engineering, architecture design, hyperparameter choices, and training strategy. Decisions are supported by evidence from the data and previous tasks.
  - 2–3 marks (Good/Satisfactory): Reasonable justification is provided for most design choices, although some decisions may lack supporting evidence or detailed explanation.
  - 1 mark (Limited): Minimal justification is provided, or design choices appear largely arbitrary.
  - 0 marks: No meaningful justification provided.
- Evaluation and interpretation of classification performance: **5 marks**
  - 4–5 marks (Excellent): Appropriate classification metrics are used and interpreted correctly. Performance is critically evaluated, strengths and limitations are identified, and conclusions are supported by evidence.
  - 2–3 marks (Good/Satisfactory): Performance is evaluated using appropriate metrics and interpreted reasonably well, although the discussion may be largely descriptive.
  - 1 mark (Limited): Evaluation is minimal, incomplete, or incorrectly interpreted.

    - 0 marks: No meaningful evaluation provided.
- Comparison with Task 3 and discussion of feature relevance: **4 marks**
    - 4 marks (Excellent): Provides a clear and critical comparison between the deep learning model and the classical machine learning model developed in Task 3. Discusses whether deep learning improves classification performance and provides a well-supported discussion of feature relevance, drawing on findings from Tasks 1, 2, and 4.
    - 2–3 marks (Good/Satisfactory): Provides a reasonable comparison with Task 3 and discusses some relevant features, although the discussion may lack depth or supporting evidence.
    - 1 mark (Limited): Only a superficial comparison is provided with limited discussion of feature relevance.
    - 0 marks: No meaningful comparison or discussion provided.

Useful links

- https://www.kaggle.com/datasets/razanihababdellatif/battery-remaining-useful-life-rul-dataset/code

Submission Instructions:

Submit your coursework as either a single Jupyter Notebook (.ipynb) or multiple Jupyter Notebooks (.ipynb) as a zip file on Tabula. All justifications, discussions, interpretations, and conclusions must be included within the PDF (or notebook(s)) using Markdown cells.

Any additional libraries required to run your notebook(s) should be listed in the first code cell together with the necessary installation commands (bash). Only packages available through PyPI (pip) or Conda may be used. The assignment should successfully run on DCS Rocky Linux system.

Ensure that your notebook(s) execute without errors, clearly identify each task, and include appropriate acknowledgement of any permitted Generative AI usage.

Students are encouraged to follow good software engineering practices, including clear code structure, meaningful variable names, appropriate commenting, and the use of code quality tools such as `ruff` to improve readability and maintainability of their code.